

Painter Naming and the Distribution of Generated Art: Feature Proximity and Variation in Two Painter Cases

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Abstract

An image generator may produce a recognizable painter reference without reproducing variation across that painter’s works. We examine how adding a painter’s name changes the feature distribution of generated paintings, using 70 digital reproductions by Monet and Cézanne and 1,006 images from three generation services. Shared scene descriptions, repeated generation and randomized prompt order support within-service comparisons. In a fixed representation of 31 color, spatial and digital-texture features, naming reduces energy discrepancy from the reference panel in all six service–painter comparisons. The four Nano Banana 2 and FLUX.2 Max contrasts reject the conditional randomization null after Holm adjustment across eight tests. Named collections have lower empirical aggregate variation than their matched artist-free collections in all 72 descriptive comparisons across feature views and processing pipelines. Decomposition shows that this does not uniformly mean less variation between repetitions of the same prompt. Proximity changes are less consistent across these views: removing texture and reducing resolution reverses one of those four primary directions. We conclude that painter naming changes both proximity and the organization of variation in these generated collections. The results concern measured digital-image distributions; reference selection, reproduction differences and absent human validation limit conclusions about artistic style.

1 Introduction

What does it mean for a generated collection to resemble a painter’s work? Individual images can evoke a familiar palette or motif while covering only a small part of the variation present across the painter’s paintings. Conversely, a collection can be diverse yet systematically differ from those paintings. Evaluating resemblance therefore requires more than distance to an average image or average feature vector. It requires comparing distributions and asking what kind of variation those distributions contain.

Painter naming offers a direct intervention with which to study this problem. Given the same description of a scene, adding “in the style of Claude Monet” changes the artistic instruction while leaving the written scene description intact. The resulting images need not change only in style: the instruction may also affect composition, depicted content or service behavior. Nevertheless, the comparison identifies how the complete generation system responds to an explicit painter reference under a common prompt design.

We study this intervention for Monet and Cézanne with two commercial model routes and a third image-generation service. The experiment uses 24 shared scene descriptions, three repetitions per description and a reference panel of 70 paintings. We measure discrepancy from each painter’s reference distribution using pairwise feature distances and measure variation within generated sets separately. A decomposition distinguishes variation between repetitions of the same description from variation between descriptions. This distinction matters: a narrower collection could result from more similar responses to different scenes, more consistent repetitions, or both.

Our contribution is an empirical account of this prompt intervention and its measurement limits. The primary comparisons show that closer feature distributions can accompany reduced

aggregate variation. Subsequent diagnostics locate that variation and test the dependence of the findings on feature choice, image processing and the reference panel. The study is a comparison of selected painter cases, rather than an estimate of how often a phenomenon occurs across art history. It also separates observable image statistics from perceptual judgments: neither proximity nor concentration alone establishes successful stylistic imitation or a loss of artistic diversity.

2 Related work

Quantitative studies of art have used color statistics, image structure and information-theoretic measures to characterize variation across painting collections (Kim et al., 2014; Sigaki et al., 2018; Lee et al., 2020). More recently, Kim et al. (2026) combined learned formal and contextual representations to study art-historical development, including a context-conditioned image-to-image experiment. Our question is different: we use text-only generation to examine how a painter-name intervention changes the distribution of interpretable features relative to a finite set of that painter’s works.

Research on generated art has also examined artistic imitation and attribution. AI-Pastiche assesses generated paintings through human authenticity and prompt-adherence surveys (Asperti et al., 2025). AI-WikiArt uses captions of original paintings to condition generation and evaluates attribution with vision-language models (Fu et al., 2025). These approaches address judgments about individual images and their attribution. Our distribution-level analysis complements them by distinguishing feature proximity from variation across and within scene descriptions; it does not replace human evaluation.

Separation between human and generated paintings in learned feature space is already documented and investigated by Asperti (2026). Consequently, scatter separation or detection accuracy alone is not the contribution here. More generally, generative-model evaluation distinguishes fidelity from diversity and coverage (Naeem et al., 2020), while preprocessing can substantially affect feature-based comparisons (Parmar et al., 2022). We apply this distinction to painter naming and use energy discrepancy (Székely and Rizzo, 2013) to compare full empirical distributions without reducing either collection to its centroid.

3 Methods

3.1 Reference paintings and generation design

The reference panel contains 38 Monet and 32 Cézanne works, represented by Wikimedia-delivered digital reproductions. The painters were selected after exploratory work on four painters; those exploratory results are not pooled into this analysis. Reference identities were screened against prior project exposure records. The available panel is not a probability sample of either painter’s oeuvre. Selection depended on image availability and landscape eligibility, with one figure-led work excluded during visual coding.

Three broad content classes organize the comparison: water-dominated scenes, built scenes and land-dominated scenes. Their reference counts are 21/4/13 for Monet and 3/11/18 for Cézanne. Coding was performed through a maintainer-run LLM visual assessment, without independent human adjudication; 17 mixed or uncertain annotations remain flagged. These classes provide coarse content adjustment. They do not match each generated image to a specific original composition.

The prompt inventory contains eight descriptions per content class. Each is used for both painters, with three repetitions, giving 72 intended images per service–painter–condition cell. The artist-free prompt begins “Create an oil painting on canvas.” The named prompt adds “In

the style of [painter].” Both then give the identical detailed scene description and request only the painting area, without a frame, signature, letters or watermark. No original painting is provided to the generator. A third, generic named condition on one service replaces the detailed description with a short scene description; this changes content specificity as well as prompt length. An exact example appears in Appendix A.

The commercial routes request Google’s `gemin-3.1-flash-image`, termed Nano Banana 2 (NB2), and Black Forest Labs’ `flux.2-max` (FLUX), through fixed OpenRouter providers. A local OAuth image service requests `gpt-image-2`, termed the OAuth service below. These are requested service identifiers, not attestations of immutable model weights. NB2 and FLUX return 1024-by-1024 images, in JPEG and PNG respectively. The OAuth service returns PNGs with variable geometry and reported quality: 424 of 430 images report low quality and six report medium, all six in artist-free conditions. Its prompt comparison therefore includes possible changes in delivered quality.

Condition order is randomized independently within each service, painter, description and repetition group. For the OAuth service, all three conditions are randomly ordered. Repetitions are distributed across eight assigned collection batches; the realized collection spans approximately 11.5 hours. Batches do not represent independent backend sessions. The 1,008 assigned slots yield 1,006 images: 288 each for NB2 and FLUX, and 430 for OAuth. Two Cézanne/OAuth/generic requests are refused and remain missing. One transient FLUX failure is recovered by an exact-payload technical retry. Successful images are retained without selecting among alternatives or screening for stylistic resemblance. Generated content classes refer to the intended prompts; adherence has not been independently assessed.

3.2 Feature representation and distribution measures

Each image is measured in 31 interpretable coordinates: 11 color, eight spatial and 12 digital-texture features (Appendix A). The primary pipeline applies orientation metadata, converts valid embedded color profiles to sRGB, assumes sRGB for compatible unprofiled files and preserves aspect ratio at a 512-pixel short side. Images are opaque and resized with Lanczos filtering, without upsampling or discretionary cropping. Each coordinate is standardized by its median and interquartile range (IQR) in a separate development collection of 221 works from four painters, with equal total weight per development painter. Neither current references nor generated images fit this transformation.

For reference vectors $X = \{x_i\}$ and generated vectors $Y = \{y_j\}$, let a_i, b_j be nonnegative weights summing to one within each set. We use the empirical V-statistic energy discrepancy

$$\begin{aligned} \widehat{E}(X, Y) = & 2 \sum_{i,j} a_i b_j \|x_i - y_j\|_2 - \sum_{i,k} a_i a_k \|x_i - x_k\|_2 \\ & - \sum_{j,l} b_j b_l \|y_j - y_l\|_2. \end{aligned} \quad (1)$$

This statistic incorporates distances within and between the two collections. Its finite-sample baseline need not be zero for independent samples from the same population. We interpret lower values as closer empirical distributions in the specified feature geometry, not as a calibrated degree of artistic similarity.

References have equal weight, $a_i = 1/N_p$. If painter p has N_{pc} reference works in class c and a generated cell has n_{pc} available images in that class, a generated image receives $b_j = N_{pc}/(N_p n_{pc})$. Thus both distributions have the reference panel’s class proportions. This weighting reduces differences in broad content mixture but leaves within-class content differences unresolved.

To measure spread, we use the weighted population-form covariance trace

$$\mathcal{V}(Y) = \sum_j b_j \|y_j - \bar{y}\|_2^2, \quad \bar{y} = \sum_j b_j y_j. \quad (2)$$

We distinguish the generated/reference ratio $\mathcal{V}(Y)/\mathcal{V}(X)$ from the named/artist-free ratio $\mathcal{V}(Y_N)/\mathcal{V}(Y_F)$. A value below one has a different comparison target in these two ratios. Coordinate-wise squared IQR sums provide a complementary spread summary.

Repeated descriptions allow an exact descriptive decomposition. For description b , let w_b be its total image weight and $\bar{y}_b$ its weighted mean. Then

$$\mathcal{V}(Y) = \underbrace{\sum_b \sum_{j \in b} b_j \|y_j - \bar{y}_b\|_2^2}_W + \underbrace{\sum_b w_b \|\bar{y}_b - \bar{y}\|_2^2}_B. \quad (3)$$

W : within descriptions B : between description means

The between-description term can be further separated into within-class and between-class components. With three repetitions per description, these are finite-set summaries, not bias-corrected estimates of population variance components. In particular, the between-description term also includes noise in means estimated from only three images.

3.3 Prompt contrasts and inference

The prespecified inferential family comprises six named-minus-artist-free energy contrasts and two detailed-minus-generic named contrasts on OAuth. A contrast uses complete pairs sharing the description, repetition and assigned batch, with the same content weights in both conditions. The paired statistic retains all terms in Equation 1; it is not a difference of average distances to a reference mean. Appendix B gives the swap identity.

For each contrast, 99,999 Monte Carlo sign assignments yield a two-sided randomization p -value with the plus-one correction. Holm adjustment controls the fixed family of eight tests at level .05. The sharp null states that prompt assignment changes neither availability nor measured features in any assigned position, under the fixed request and technical-retry policy and no interference. For a pairwise OAuth comparison, the third condition’s position is conditioned on. The tests concern this finite design and reference panel. They do not test an oeuvre-wide mean effect, perceptual equivalence or equality of original and generated population distributions. Treatment-dependent call duration or service carryover could violate no interference.

The prompt inventory, primary metrics and test family were specified before current evaluation features were measured. Feature-view, variance-decomposition, reference-influence, painter-interaction and additional detection/coverage analyses were designed after the primary results were observed. These subsequent analyses are descriptive; they introduce no additional significance tests or population confidence intervals.

3.4 Measurement and reference diagnostics

We examine three processing pipelines: the primary 512-pixel pipeline, a 256-pixel version and common JPEG encoding at quality 90 with 4:2:0 subsampling at 512 pixels. Each uses its own development-only scaler. Four feature views cross these pipelines: all 31 coordinates; 28 coordinates after dropping three derived multiscale summaries; 31 coordinates divided by the square root of their family size; and the 19 color/spatial coordinates with texture removed. The family-rescaled squared distance sums mean squared differences within each family; it does not equalize observed family variance or perceptual importance. This gives 12 views per service-painter comparison.

Reference diagnostics delete one work, holding-collection membership group or content class at a time. Collection groups use exact recorded sets of holding- collection IDs and do not identify photographic capture workflows. Work and collection-group deletions preserve the original content-class masses while reweighting the remaining works within each class; class deletion renormalizes the surviving class masses. A separate analysis refits the scaler after omitting

each development painter. Painter specificity uses equal one-third content-class weights for both painters. With G_M, G_C denoting collections assigned to Monet and Cézanne, define

$$I = \widehat{\mathcal{E}}(X_M, G_M) + \widehat{\mathcal{E}}(X_C, G_C) - \widehat{\mathcal{E}}(X_M, G_C) - \widehat{\mathcal{E}}(X_C, G_M). \quad (4)$$

Negative I favors own-painter over cross-painter alignment. The within-reference and within-generated terms cancel, making this a contrast of relative alignment. Artist-free prompts are identical across the two collection labels and provide a descriptive control for their arbitrary painter assignment.

For visualization, we fit one common principal-component analysis (PCA) per painter, assigning half the weight to references and sharing half equally across the seven generated cells, with content weighting within each. All cells use that painter’s common axes. Full-space linear and radial-basis-function (RBF) kernel-ridge classifiers use six folds holding out entire generated descriptions and disjoint original works, without tuning or PCA inputs. This grouping respects the repeated-prompt structure (Roberts et al., 2017). Cross-service evaluation trains on one service and tests on another using the same held-out description/work scheme. We report balanced accuracy at a fixed zero decision threshold and mean within-fold AUC separately; the former also depends on calibration. Technical classifier and coverage details appear in Appendix C.

4 Results

4.1 Painter naming improves primary feature proximity

Painter naming lowers primary energy discrepancy in all six service–painter comparisons (Figure 1). NB2 decreases from 4.188 to 3.342 for Monet and from 4.704 to 2.886 for Cézanne; FLUX decreases from 1.990 to 1.365 and from 1.700 to 0.804, respectively. These four contrasts reject the conditional sharp null after adjustment across the eight-test family (Table 1). The OAuth differences are smaller, and neither survives adjustment. The detailed-versus-generic comparisons likewise provide no rejection after adjustment. Non-rejection does not establish equivalence.

The same primary feature space also shows that the named collections have less aggregate spread than the references. Generated/reference trace ratios for the six detailed named cells range from .354 to .756, whereas the four NB2 and FLUX artist-free cells range from 1.069 to 1.535. Thus, reduced spread is not a general property of all generated images in this experiment. It is associated with the tested painter-naming conditions. The two generic named OAuth cells have still lower reference-relative ratios, .306 for Monet and .395 for Cézanne, although their energy contrasts do not establish an advantage of detailed prompting.

Comparison	Service	Painter	Pairs	$\Delta\widehat{\mathcal{E}}$	Holm p
Named – free	NB2	Monet	72	−0.846	.00310
	NB2	Cézanne	72	−1.818	.00008
	FLUX	Monet	72	−0.625	.00008
	FLUX	Cézanne	72	−0.896	.00008
	OAuth	Monet	72	−0.234	.19308
	OAuth	Cézanne	72	−0.056	1.00000
Detailed – generic	OAuth	Monet	72	−0.037	1.00000
	OAuth	Cézanne	70	−0.272	.09280

Table 1: The complete prespecified test family, using the primary 31-feature representation and matched pairs. Negative changes indicate smaller energy in the first condition. Values are conditional randomization results, with 99,999 sign draws and Holm correction across all eight comparisons. Both conditions in the final two rows name the painter.

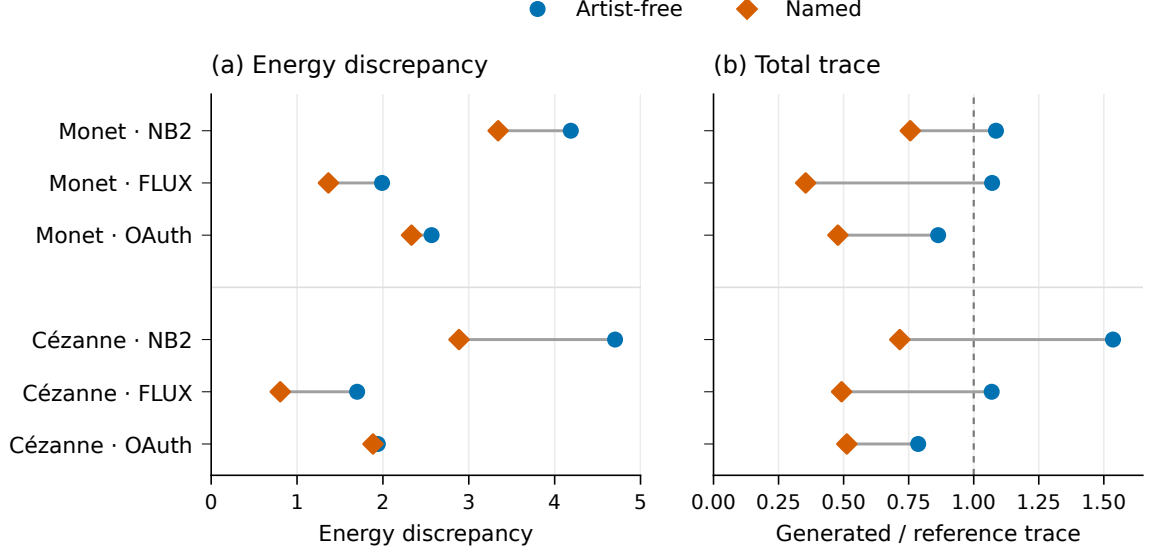


Figure 1: Primary feature discrepancy and spread under artist-free and named prompting. Energy compares each generated distribution with its painter’s reference panel. Trace ratios compare generated spread with reference spread; the line at one denotes equal empirical trace. Each displayed generated cell contains 72 images; reference panels contain 38 Monet and 32 Cézanne works. The panels are descriptive and do not share a numerical scale.

Energy and spread are distinct summaries but are not statistically independent: Equation 1 includes a within-generated distance term. These results therefore describe the joint behavior of two measurements, not evidence for a necessary trade-off between fidelity and diversity.

Under equal content-class weights, naming also strengthens relative own-painter alignment. The interaction I changes from .101 to $-.855$ for NB2, from $-.048$ to -1.755 for FLUX and from $-.044$ to -1.802 for OAuth. Its named-minus-free change is negative in all 36 service/view/pipeline combinations. This joint contrast does not require each generated collection individually to be closest to its own painter’s references. Separately generated artist-free collections have identical prompt payloads across painter labels yet nonzero empirical discrepancies. These controls illustrate finite-set and collection variability; they do not provide an equivalence threshold.

4.2 Aggregate contraction has different within-prompt components

The named/artist-free total-trace ratios range from .331 to .697 in the primary representation. Decomposition reveals different sources of this contraction (Figure 2). FLUX reduces both within-description and between-description variation strongly: the respective ratios are .309 and .337 for Monet, and .489 and .451 for Cézanne. For NB2/Cézanne, within-description variation decreases to .786 while between-description variation decreases to .378. Its aggregate reduction therefore conceals a much larger relative change between descriptions than between repetitions of a description.

The OAuth/Cézanne case shows why the distinction is necessary. Total trace falls to .651 of its artist-free value even though within-description trace rises to 1.139; between-description trace falls to .583. A tighter overall cloud can therefore coexist with more variable repetitions. These observations do not identify a latent model mechanism or demonstrate fewer perceptually distinct outputs. They locate the measured variation in the actual repeated-prompt design.

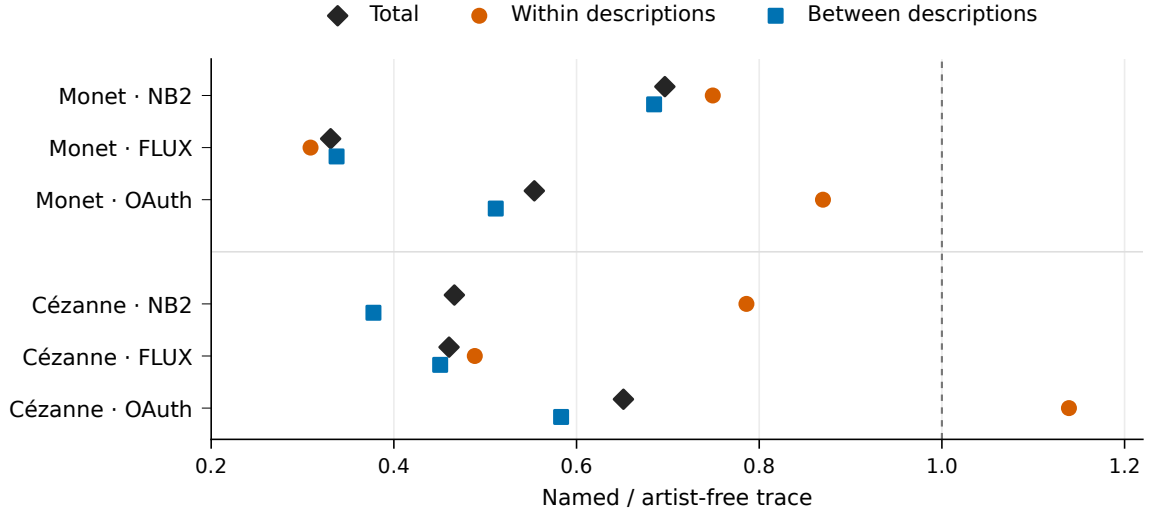


Figure 2: Where generated variation changes under naming. Each point is a named/artist-free trace ratio in the primary representation. Total variation is the sum of within-description and between-description components, so its ratio need not equal their unweighted average. The line at one denotes no change. OAuth/Cézanne contracts in total while increasing within-description variation. These finite-set decompositions are descriptive.

4.3 Contraction is more consistent than proximity across feature views

All 72 named/artist-free total-trace ratios across the four feature views and three pipelines are below one, ranging from .308 to .816. Proximity is less consistent: 66 of 72 energy changes are negative. Both FLUX painter comparisons and NB2/Cézanne remain negative in all 12 view/pipeline combinations. For NB2/Monet, the 256-pixel color/spatial-only comparison instead increases energy by .416. The same no-texture view at 512 pixels gives a named/reference trace ratio of 1.101. Consequently, robust contraction relative to artist-free outputs does not imply robust below-reference spread or uniformly improved proximity.

The sensitivity has a substantive measurement basis. Texture coordinates contribute 49.0% and 47.1% of primary reference trace for Monet and Cézanne. Some multiscale coordinates are strongly correlated in development data. Agreement among such coordinates does not constitute independent evidence of the same artistic property. The diagnostic views probe this dependence without selecting a preferred representation from the observed effects.

Deleting individual reference works, holding-collection groups or content classes preserves the direction of all four NB2/FLUX primary proximity changes. These directions also survive omission of any one development painter from scaling. Some OAuth changes cross zero under these perturbations. Deletion stability supports the paid-service comparisons within the available panel, but does not correct shared reproduction differences or turn that panel into an oeuvre sample.

4.4 Visible overlap and detectability answer different questions

The common PCA views show overlapping clouds with changes in location and concentration (Figure 3). The first two components retain 48.4% of weighted variation for Monet and 46.0% for Cézanne, leaving more than half outside the plots. Visual overlap consequently cannot establish equality of the full distributions. Conversely, separation in a projection would not establish a difference in artistic style.

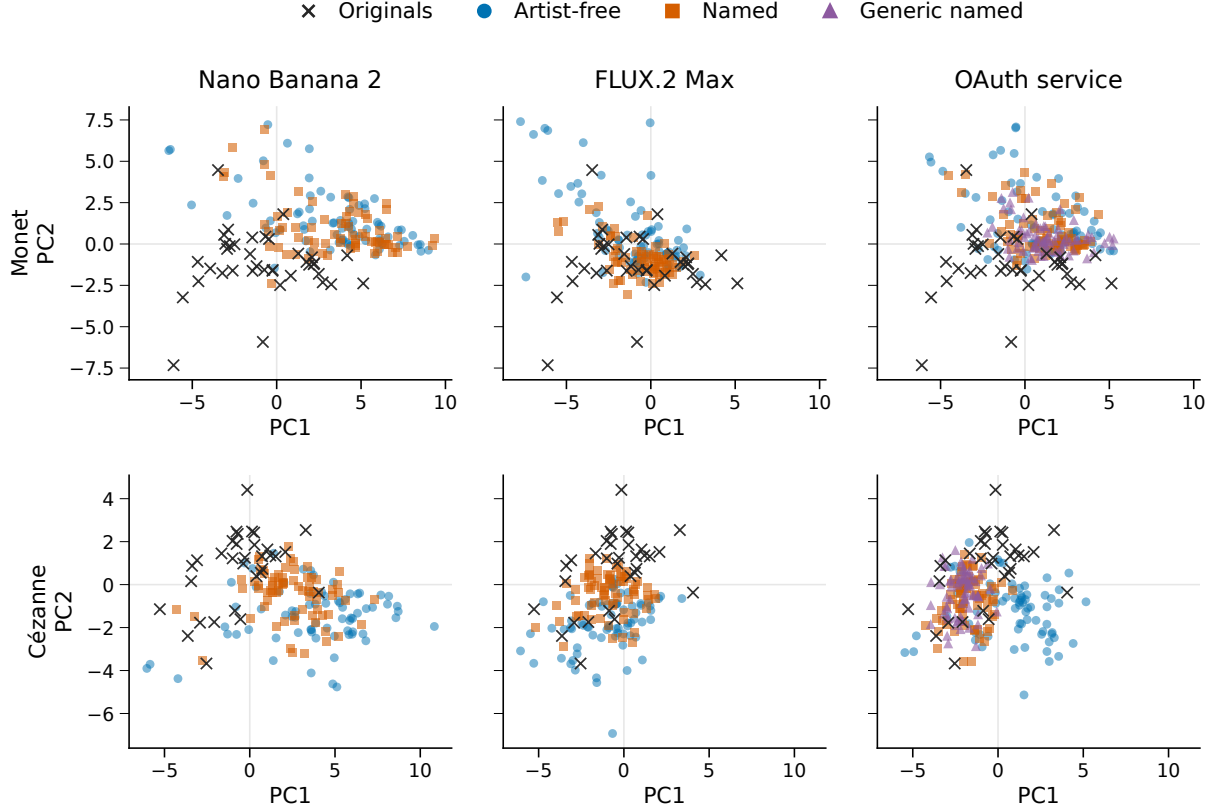


Figure 3: Original and generated feature distributions in common painter-specific PCA coordinates. Every image is shown; references are repeated across service panels for comparison. Axes and limits are shared within each painter, but the two painters have separate PCA fits. The generic named condition is available only for OAuth. Projections show overlap and concentration while retaining less than half of total weighted variation.

In the full feature space, within-service RBF balanced accuracy ranges from .893 to .987 across the 14 original/generated comparisons. However, all 576 NB2/FLUX images are square and all 70 references are nonsquare: a rule using only square geometry separates these domains perfectly. Capture workflows for the reference reproductions are also unresolved. High classification accuracy therefore establishes differences between the recorded image domains, not a validated stylistic distinction.

Cross-service behavior further limits detector interpretation. At the fixed threshold, naming lowers balanced accuracy in every directed service/painter comparison for both kernels. Named-condition ranges are .410–.874 for the linear kernel and .447–.704 for RBF. This need not mean uniformly weaker ranking discrimination. For Monet with training on FLUX and testing on OAuth, linear balanced accuracy falls from .904 to .874 while mean within-fold AUC rises from .953 to .968. The observed threshold transfer includes calibration changes; below-chance scores are not evidence that generated images have become more authentic.

5 Discussion

The central result is that painter naming changes proximity and variation together, but the two measurements require separate interpretation. In the primary representation, NB2 and FLUX generate distributions closer to their painter references under naming, accompanied by lower total variation. The repeated-description design shows why describing this simply as “diversity

loss” is insufficient. Naming can reduce differences between responses to different descriptions while leaving variability between repetitions relatively intact, or even increasing it. Evaluations that pool all generated images into one cloud conceal this distinction.

One possible explanation is that painter instructions encourage recurrent visual choices across otherwise different scenes. Our measurements are compatible with that interpretation but do not distinguish it from changes in depicted content, composition or rendering. In particular, broad prompt classes do not guarantee matched realized scenes. Human judgments of set-level stylistic range and independent content coding would be needed to test whether the observed changes correspond to stylistic concentration as perceived by viewers.

The representation also determines what “closer” means. Interpretable coordinates make color, spatial and texture contributions inspectable, but they do not provide a complete or independently validated model of style. Texture has substantial weight in the current feature geometry, and the NB2/Monet counterexample prevents a representation-independent proximity claim. The 72 sensitivity comparisons reuse the same images and are not 72 independent replications. Robustness across their fixed views is evidence about measurement choices, not about new painters, content inventories or generation runs.

Reference quality constrains the absolute comparison more strongly than image count alone. The current panel contains selected, unequally represented content classes and digital reproductions with unresolved capture workflows. Shared aspect ratio within the paid named/free comparisons removes a simple binary geometry difference between those prompt arms, but not the original/generated geometry mismatch or other rendering and content effects. Collection duration, mutable service implementations and the OAuth quality imbalance further limit trans-
portability. None of these results supplies a universal model ranking.

The next validation should therefore use independently coded reference panels, controlled reproductions and human judgments of the same generated sets. A learned representation could test another feature geometry, but agreement with it would not by itself establish perceptual validity. Additional generations are most informative when they replicate the intervention with new descriptions, painter cases or independently repeated collections, rather than merely enlarge the current cells.

6 Conclusion

For these Monet and Cézanne reference panels, painter naming often improves primary feature proximity while reducing aggregate generated variation. Within the fixed diagnostic grid, aggregate contraction is more consistent across measurement variants than proximity improvement, and it does not uniformly reflect reduced variation between repetitions. Distribution-based evaluation of generated art should therefore report what changes within and between prompts, alongside its reference and representation choices. The present results establish a bounded computational finding, not an assessment of artistic worth or perceptual style reproduction.

Data and code availability

The accompanying repository contains the exact prompt inventory, reference identities and annotations, standardized feature vectors, analysis code, complete numeric tables and figure-generation code. The controlled-study and subsequent diagnostic reports retain all prespecified endpoints and the additional sensitivity results summarized here. The manuscript README gives their locations and offline reproduction commands. Raw painting reproductions and full-resolution generated images are retained separately and are not included in the repository’s redistributed files. Numeric replay does not require image-generation API access.

A Prompt and feature specification

An example detailed named prompt is:

Create an oil painting on canvas. In the style of Claude Monet. Scene: A pond viewed from close to its edge, with groups of floating lily pads, gaps of open water and reflections of surrounding trees. The water fills the scene; no horizon or prominent person is visible. Render only the painting area, without a surrounding frame, signature, letters or watermark.

The artist-free version removes only the painter-style sentence. The generic named version substitutes “A pond with water lilies.” for the detailed scene. All 24 descriptions are available in the accompanying prompt inventory.

Family	Count	Coordinates
Color	11	Lightness median/IQR; chroma median/IQR; chromatic fraction; hue concentration/entropy; chromatic differences at three scales and their slope.
Spatial	8	Spectral slope/residual/anisotropy; orientation entropy; horizontal-vertical balance; gradient median/IQR; quadrant divergence.
Digital texture	12	Four wavelet energies, their slope and curvature; three local-binary-pattern entropies; local coefficients of variation at three scales.

Table 2: The fixed interpretable representation. Coordinates describe digital image statistics, not physical paint relief. The 28-coordinate view removes chromatic-difference slope, wavelet slope and wavelet curvature.

B Paired randomization statistic

Let A_i, B_i be a retained pair with common weight b_i , and define $C(v) = \sum_j a_j \|v - x_j\|_2$ and $T(v) = \sum_i b_i (\|v - A_i\|_2 + \|v - B_i\|_2)$. Then

$$c_i = b_i \{2[C(B_i) - C(A_i)] - [T(B_i) - T(A_i)]\}, \quad \sum_i c_i = \hat{\mathcal{E}}(X, B) - \hat{\mathcal{E}}(X, A). \quad (5)$$

Complementary pair swaps produce $\sum_i s_i c_i$, $s_i \in \{-1, 1\}$, exactly, including the within-generated distances. Independent randomized condition positions justify the sign assignments under the stated sharp null. Repeated scene descriptions are not treated as independently sampled scenes for population inference. With e sign draws at least as extreme in absolute value as the observed statistic, including ties, $p = (1 + e)/100,000$. All contrasts retain 24 distinct descriptions; the two missing generic outputs reduce only the Cézanne detailed/generic comparison to 70 pairs.

C Supporting distribution diagnostics

For $d = 31$ standardized coordinates, the fixed kernels are $K_L(u, v) = u^\top v / d + 1$ and $K_R(u, v) = \exp(-\|u - v\|_2^2 / d) + 1$. The classifiers minimize weighted squared error with a kernel penalty of .01, using equal total training weight for original and generated labels and the reference content proportions within each. Training and evaluation memberships are fixed. Sparse reference classes mean not every test fold contains all three classes; pooled balanced accuracy uses the complete held-out set. For cross-service evaluation, AUC is computed within each fold and then averaged across six folds, rather than pooling margins from differently trained classifiers. Complete kernels, memberships and scores are included with the analysis code and numeric tables.

Reference coverage assigns each original a ball whose radius is the distance to its k th nearest distinct original, and measures the reference mass with at least one generated neighbor inside that ball. We examine $k = 1, 3, 5$ and both uniform and reference-class-matched generated subsamples, without replacement. For each setting, 100 draws use either N_p or $\lfloor N_p/2 \rfloor$ generated images; half-size class counts use largest-remainder allocation. The operating point matters: FLUX/Cézanne/named coverage is .438, .750 and .969 at these three neighborhood sizes with all generated images available. For OAuth/Monet, the named-minus-free direction at $k = 3$ changes from negative with all generated images to positive for median class-matched subsample coverage. Thus coverage is not an invariant measure of how much of a painter’s style has been reproduced.

Disjoint reference subsets provide a finite-panel real/real control. Each draw allocates $\lfloor N_{pc}/2 \rfloor$ works per class to each subset, giving 18 Monet or 15 Cézanne works per side; these draws reuse the same panel. At $k = 3$, median coverage of reduced reference anchors is .944 and .933 for held-out real queries, compared with .833 and .800 for FLUX named queries at matched sizes. At $k = 5$, both real controls and the Cézanne FLUX named control reach median coverage one. Such saturation does not establish distributional equivalence, and repeated subsets do not supply independent reference replications.

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